

SIMATS ENGINEERING

COURSE

NAME: Cryptography and Network Security

COURSE CODE: CSA51

ASSESSMENT TOOL 1

: -

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I J.Vishnu Vardhan(192324297) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

QUESTION 3 – SECURE MESSAGING SYSTEM

Scenario

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word **HELLO** using the keyword **KEY** to ensure confidentiality.

Questions

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

INTRODUCTION AND THEORY

1. Aim

To demonstrate the encryption of the plaintext **HELLO** using the **Vigenère Cipher** with the keyword **KEY**, and to explain how the repeating keyword influences each character of the resulting ciphertext.

2. Objectives

- 1. To understand the working principle of the Vigenère Cipher.
- 2. To convert alphabetic characters into numerical values.
- 3. To apply the encryption formula.
- 4. To understand the role of a repeating keyword.
- 5. To generate the ciphertext from the given plaintext.
- 6. To analyze how the Vigenère Cipher differs from the Caesar Cipher.

3. Given Information

Parameter	Value
Plaintext	HELLO
Keyword	KEY
Encryption Technique	Vigenère Cipher
Alphabet	English A–Z
Number of letters	26

The standard numerical representation is: **A = 0, B = 1, C = 2, ..., Z = 25.**

4. Introduction to Vigenère Cipher

The **Vigenère Cipher** is a classical encryption technique based on substitution. It uses a **keyword** to determine the shift applied to each plaintext character. Unlike the Caesar Cipher, which uses one fixed numerical shift, the Vigenère Cipher uses different shifts according to the letters of the keyword.

5. Encryption Formula

$C = (P + K) \bmod 26$

Where C = ciphertext value, P = plaintext value, K= keyword value, and 26 = number of letters in the alphabet.

6. Basic Working Principle

Plaintext ☐ Convert letters to numbers ☐ Repeat keyword ☐ Convert keyword to numbers ☐
Apply $C = (P + K) \bmod 26$ ☐ Convert numbers back to letters ☐ Ciphertext

PAGE 2 – STEP-BY-STEP ENCRYPTION

Step 1 – Write the Plaintext

The given plaintext is **HELLO**.

H E L L O

There are 5 characters.

Step 2 – Write the Keyword

The given keyword is **KEY**. Since the plaintext contains 5 characters, the keyword must be repeated until it also contains 5 characters.

KEY □ **KEYKE**

Plaintext : H E L L O Keyword : K E Y
K E

Step 3 – Convert Plaintext to Numbers

Using A = 0, B = 1, C = 2, ..., Z = 25:

H = 7

E = 4

L = 11

L = 11

O = 14

Step 4 – Convert Keyword to Numbers

The repeated keyword is **KEYKE**: K = 10, E = 4, Y = 24, K = 10, E = 4.

Complete Initial Table

Position	Plaintext	P	Keyword	K
1	H	7	K	10
2	E	4	E	4
3	L	11	Y	24
4	L	11	K	10
5	O	14	E	4

CALCULATIONS AND RESULT

7. Character-by-Character Encryption

Character 1: H = 7, K = 10

$$C = (7 + 10) \bmod 26 = 17 \square R$$

$$H + K = R$$

Character 2: E = 4, E = 4

$$C = (4 + 4) \bmod 26 = 8 \square I$$

$$E + E = I$$

Character 3: L = 11, Y = 24

$$C = (11 + 24) \bmod 26 = 35 \bmod 26 = 9 \square J$$

$$L + Y = J$$

Character 4: L = 11, K = 10

$$C = (11 + 10) \bmod 26 = 21 \square V$$

$$L + K = V$$

Character 5: O = 14, E = 4

$$C = (14 + 4) \bmod 26 = 18 \square S$$

$$O + E = S$$

8. Final Encryption Table

Pos.	Plain	P	Key	K	Calculation	C	Cipher
1	H	7	K	10	$(7+10) \bmod 26$	17	R
2	E	4	E	4	$(4+4) \bmod 26$	8	I
3	L	11	Y	24	$(11+24) \bmod 26$	9	J
4	L	11	K	10	$(11+10) \bmod 26$	21	V
5	O	14	E	4	$(14+4) \bmod 26$	18	S

Therefore: HELLO $\square$ RIJVS

9. How the Repeating Keyword Influences Each Character

The keyword is KEY and is repeated as KEYKE. Each keyword character determines the shift for the corresponding plaintext character.

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E
Shift	10	4	24	10	4
Ciphertext	R	I	J	V	S

ANALYSIS AND CONCLUSION

10. Why is the Keyword Repeated?

The keyword KEY contains three letters, whereas HELLO contains five letters. Therefore, the keyword is repeated as KEYKE. If the plaintext were longer, the same process would continue until every plaintext character had a corresponding key character.

11. Comparison with Caesar Cipher

Feature	Caesar Cipher	Vigenère Cipher
Key	Single shift	Keyword
Number of shifts	One	Multiple
Example	Shift +3	KEY
Repeating key	Not required	Required
Security	Relatively weak	Stronger than Caesar
Encryption	Same shift	Different shifts

12. Advantages

1. Uses multiple substitution shifts.
2. More secure than a simple Caesar Cipher.
3. Uses a keyword instead of a single number.
4. Easy to understand and implement.
5. Provides different ciphertext characters for similar plaintext characters in many cases.

13. Limitations

1. It is a classical cipher and is not secure for modern confidential communication.
2. Repeating keywords can create patterns.
3. A weak or short keyword can make cryptanalysis easier.
4. Modern computers can break classical ciphers with suitable analysis.
5. It does not provide modern authentication or integrity protection.

14. Applications

Historically, the Vigenère Cipher was used for confidential written communication, military communication, and classical cryptography studies. Today it is mainly used for learning cryptographic concepts.

15. Result

Plaintext = HELLO
Keyword = KEY
Ciphertext = RIJVS

16. Conclusion

The Vigenère Cipher successfully transforms HELLO into RIJVS using the keyword KEY. The keyword is repeated as KEYKE, and each keyword character determines the shift applied to the corresponding plaintext character. This demonstrates the main advantage of the Vigenère Cipher over the Caesar Cipher: it uses multiple shifting values instead of one fixed shift.

QUESTION 5 – LEGACY ENCRYPTION IN CORPORATE COMMUNICATION

Scenario

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. The received message appears as: **WECRLTEERDSOEFEAOCAIVDEN**.

Questions

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages.

INTRODUCTION AND THEORY

1. Aim

To decrypt the given ciphertext using a **3-Rail Rail Fence Cipher**, determine the original message, analyze its security weaknesses, and propose a practical enhancement for secure corporate communication.

2. Objectives

- To understand the Rail Fence transposition cipher.
- To identify the meaning of a three-level arrangement.
- To decrypt the given ciphertext.
- To verify the decrypted plaintext.
- To analyze the weaknesses of the technique.
- To recommend a modern security enhancement.

3. Given Data

Parameter	Value
Ciphertext	WECRLTEERDSOEFEAOCAIVDEN
Number of rails	3
Cipher type	Transposition Cipher
Technique	3-Rail Rail Fence Cipher

4. Introduction to Transposition Cipher

A transposition cipher is an encryption technique in which the original characters are rearranged according to a particular pattern. Unlike a substitution cipher, the characters are not replaced; their positions are changed.

5. Rail Fence Cipher

The Rail Fence Cipher is a type of transposition cipher. In a 3-rail Rail Fence Cipher, the plaintext is written in a zigzag pattern across three rows or rails. The ciphertext is obtained by reading the rows from top to bottom.

6. Three-Rail Pattern

Rail 1 □ Rail 2 □ Rail 3 □ Rail 2 □ Rail 1 □ ...

DECRYPTION PROCESS

7. Given Ciphertext

The ciphertext is **WECRLTEERDSOEFEAOCAIVDEN**. It contains **24 characters**. The number of rails is **3**.

8. Determine the Rail Pattern

For 3 rails, the pattern repeats every $2(n - 1)$ characters. With $n = 3$: $2(3 - 1) = 4$. Therefore, the pattern repeats every four characters: Rail 1 → Rail 2 → Rail 3 → Rail 2.

9. Position and Rail Table

Position	1	2	3	4	5	6	7	8	9	10	11	12
Rail	1	2	3	2	1	2	3	2	1	2	3	2
Position	13	14	15	16	17	18	19	20	21	22	23	24
Rail	1	2	3	2	1	2	3	2	1	2	3	2

Therefore, Rail 1 contains 6 characters, Rail 2 contains 12 characters, and Rail 3 contains 6 characters.

10. Divide the Ciphertext

Rail 1 – First 6 characters: **WECRLT**

Rail 2 – Next 12 characters: **EERDSOEFEAO**

Rail 3 – Last 6 characters: **CAIVDN**

The rails are therefore:

Rail 1: W E C R L T

Rail 2: E E R D S O E E F E A O Rail 3: C A I V D

N

RECONSTRUCTION AND VERIFICATION

11. Arrange the Rails

Rail 1: W E C R L T

Rail 2: E E R D S O E E F E A O Rail 3: C A I V D

N

To decrypt, characters are read following the zigzag pattern: Rail 1 □ Rail 2 □ Rail 3 □ Rail 2, and the pattern is repeated.

12. Step-by-Step Reconstruction

The characters are selected according to the rail sequence. Continuing this process across all 24 positions reconstructs the original plaintext:

WEAREDISCOVEREDFLEEATONCE

Adding spaces gives:

WE ARE DISCOVERED FLEE AT ONCE

13. Original Message

The most probable original message is **WE ARE DISCOVERED FLEE AT ONCE**.

14. Verification

Take the plaintext WE ARE DISCOVERED FLEE AT ONCE and remove spaces: WEAREDISCOVEREDFLEEATONCE. Arrange it in a three-rail zigzag and read each rail horizontally.

The resulting rails are:

Rail 1: **WECRLTE**

Rail 2: **ERDSOEFEAO**

Rail 3: **CAIVDN**

Combining the rails produces the given ciphertext: **WECRLTEERDSOEFEAOCAIVDEN**. Thus, the decryption is verified.

15. Answer for Part (a)

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN Plaintext:
WE ARE DISCOVERED FLEE AT ONCE

SECURITY ANALYSIS AND ENHANCEMENT

16. Part (b) – Why the Method is Vulnerable

- 1. It only rearranges characters:** The cipher does not change the actual letters; it only changes their positions.
- 2. Very small key space:** The main parameter is the number of rails. An attacker can try 2, 3, 4, 5 rails and so on until meaningful text is found.
- 3. Frequency information is preserved:** Because letters are only rearranged, their frequencies remain unchanged.
- 4. Vulnerable to brute force:** An attacker can quickly test different rail numbers, especially for short messages.
- 5. No authentication:** The cipher does not verify who sent the message.
- 6. No integrity protection:** The receiver cannot reliably determine whether the ciphertext was modified during transmission.
- 7. Not suitable for modern corporate communication:** It is inadequate for passwords, financial information, customer information, confidential plans, and sensitive business documents.

17. Security Evaluation

Criterion	Rail Fence Cipher
Cipher type	Transposition
Number of rails	3
Changes character values?	No
Changes character positions?	Yes
Key space	Small
Brute-force resistance	Very low
Frequency protection	Poor
Authentication	No
Integrity	No
Modern security	Very weak

18. Part (c) – Practical Enhancement

A practical enhancement is to replace the Rail Fence Cipher with a modern authenticated encryption algorithm such as **AES-GCM**.

Improved system:

Original Message ☐ AES-GCM Encryption ☐ Ciphertext + Authentication Tag ☐ Secure Network ☐ AES-GCM Decryption ☐ Original Message

19. Advantages of AES-GCM

Confidentiality: Unauthorized people cannot easily understand the encrypted message.

Integrity: If the encrypted message is modified, the receiver can detect the modification.

Authentication: The authentication tag helps verify that the ciphertext was not altered.

Modern Security: AES-GCM is designed for modern computer and network communication and is far

stronger than classical transposition methods.

20. Comparison of Old and Improved Methods

Feature	Rail Fence	AES-GCM
Encryption type	Transposition	Authenticated encryption
Security	Very low	Strong
Character rearrangement	Yes	No
Modern cryptographic security	No	Yes
Integrity	No	Yes
Authentication	No	Yes
Suitable for sensitive data	No	Yes
Recommended today	No	Yes

21. Result

The ciphertext **WECRLTEERDSOEEFEAOCAIVDEN** was successfully decrypted using the 3-Rail Rail Fence Cipher. The original plaintext is **WE ARE DISCOVERED FLEE AT ONCE**.

22. Conclusion

The experiment demonstrates the working of a 3-Rail Rail Fence Transposition Cipher. The given ciphertext was successfully reversed to obtain the original message. Although the Rail Fence Cipher is useful for understanding classical cryptography, it is not secure for modern corporate communication because it only rearranges characters, has a very small key space, preserves character frequencies, and can be easily attacked. For practical corporate communication, the legacy method should be replaced with a modern authenticated encryption algorithm such as AES-GCM, which provides confidentiality, integrity, and authentication.